End-Term Examination

(CBCS)(B. Tech CSE_AI SUBJECTIVE TYPE)

(April, 2022)

Subject Code: BAS 109	**Subject: Applied Mathematics**

Time : 3 Hours **Maximum Marks : 60**

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1.		**(5*4=20)**
	(a) If λ is a characteristic root of a non – singular matrix A, then prove that $\lvert A\rvert/\lambda$ is a characteristic root of the matrix $Adj\ A$.	
	(b) Define a subspace of a vector space. Show that the set: $W = \{(a_1, a_2, a_3): a_1 + a_2 + a_3 = 0\ ,\ a_1, a_2, a_3 \in R\}$ is a subspace over the field of real numbers.	
	(c) If $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + \frac{1}{2}\cot u = 0$.	
	(d) Sketch the region of integration for the integral $\int_{-1}^{2}\int_{x^2}^{x+2} f(x,y)\,dy\,dx$ and write an equivalent integral with the order of integration reversed.	
	UNIT-1	
Q2.	(a) Define the rank of a matrix. Reduce the matrix $A = \begin{matrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{matrix}$ to the normal form by means of elementary operations and hence, determine its rank.	**(7+3)**
	(b) Find the characteristic roots of the matrix $A = \begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix}$	
Q3.	(a) State Cayley Hamilton Theorem. Use Cayley Hamilton Theorem to find the inverse of the matrix $A = \begin{vmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{vmatrix}$	**(5+5)**
	(b) For what values of λ and μ, do the following system of equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ have (i) a unique solution (ii) no solution (iii) an infinite number of solutions.	
	UNIT-2	
Q4.	(a) Define a basis and dimension of a vector space over a field F. Let $V = R^4(R)$ and $W = \{(a,b,c,d) \in R^4: a = b + c, c = b + d\}$ be a subspace of V. Exhibit a basis of W. What is $dim_R W$? (b) Use the method of Laplace transform to solve the initial value problem $y'' + 9y = 18t, y(0) = 0, y'(0) = 6$.	**(5+5)**
Q5.	(a) Let $T: R^2 \to R^2$ be a linear transformation such that $T(1,1) = (1,3)$, $T(-1,1) = (3,1)$. Find $T(a,b)$ for any $(a,b) \in R^2$. (b) Use the convolution theorem to find the inverse Laplace transform of $F(s) = \frac{1}{(s^2+a^2)^2}$	**(5+5)**
	UNIT-3	
Q6.	(a) Let $f(x,y) = \frac{x^3y}{x^2+y^2}$ $,x^2 + y^2 \neq 0, f(0,0) = 0$. Prove or disprove $f_{xy}(0,0) = f_{yx}(0,0)$. (b) If $y = \sin(m\sin^{-1}x)$, show that $(1-x^2)y_{n+2} = (2n+1)\,x\,y_{n+1} +$	**(6+4)**

	$(n^2 - m^2)y_n = 0.$	
Q7.	(a) Using the method of Lagrange's multiplier, find the maximum and minimum values of the function $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$. (b) Find the *nth* derivative of $y = (ax + b)^{-1}$.	**(6+4)**
	UNIT-4	
Q8.	(a) A vector field $\vec{F}$ is given by $\vec{F} = \sin y\,\hat{\imath} + x(1 + \cos y)\,\hat{\jmath}$. Using Green's theorem, evaluate the line integral $\int \vec{F}.d\vec{r}$ taken along the closed circular path given by $x^2 + y^2 = a^2$. (b) Find the area bounded by the curves $y = \sin x, y = \cos x$ and the line $x = 0$.	**(5+5)**
Q9.	(a) Evaluate $\int[(x + y)dx + (y - x)dy)]$ along the parabola $y^2 = x$ from $(1, 1)$ to $(4, 2)$. (b) Evaluate $\int_0^3 \int_0^4 \int_{x=y/2}^{x=(\frac{y}{2})+1} \left(\frac{2x-y}{2} + \frac{z}{3}\right) dx\,dy\,dz$ by applying the transformation $u = \frac{2x-y}{2}, v = \frac{y}{2}, w = \frac{z}{3}$ and integrating over an appropriate region in uvw- space.	**(5+5)**